

Continuous processing of biopharmaceutical proteins

Be in or be out?

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Content

- What do we mean by continuous processing?
- What happened in other industries, and in ours?
- How do we judge new technology?
- Upstream processing aspects
- Downstream processing aspects
- Economic considerations
- Conclusions: Are you in or are you out?

Continuous processing

Definitions

Continuous Process

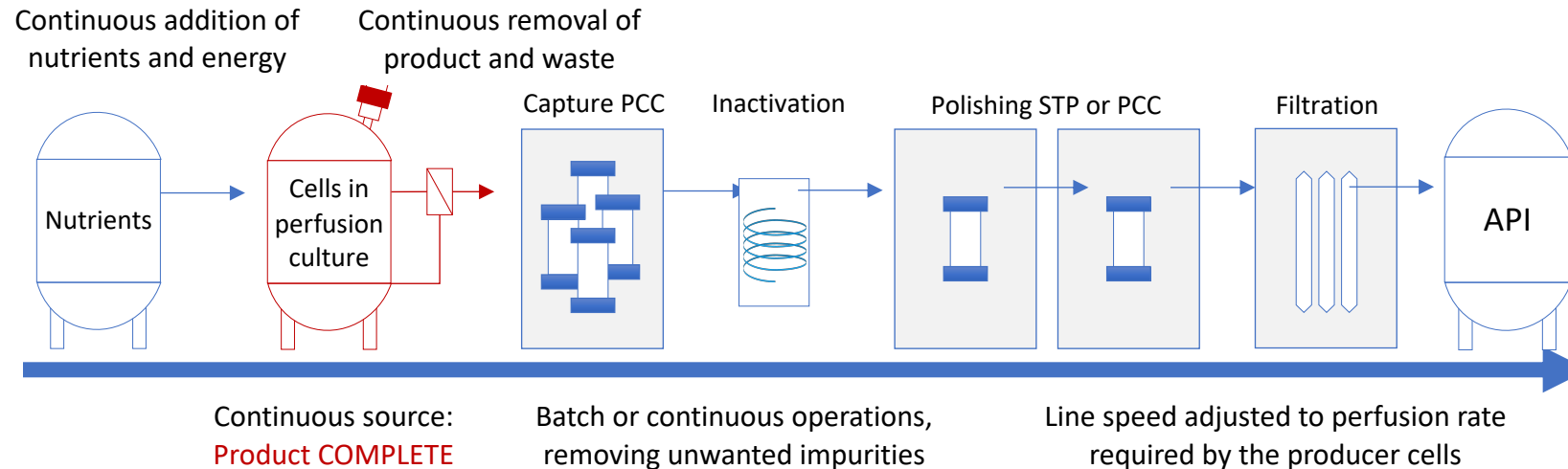
- A streamlined process that involves ongoing production of end products or services. This process keeps going on 24 hours a day and 7 days a week.
- Continuous production is a flow production method used to manufacture, produce, or process materials without interruption. The materials, either dry bulk or fluids that are being processed are continuously in motion, undergoing chemical reactions or subject to mechanical or heat treatment.

Steady State:

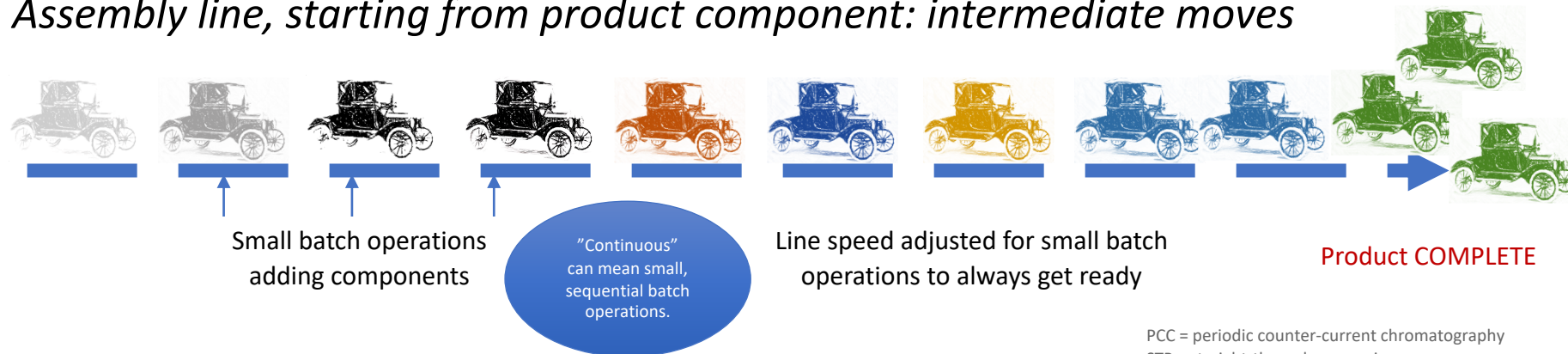
- Steady state may refer to an operating condition in thermodynamic and other systems or processes when variables stay constant as time passes. For an entire system to be at steady state, i.e. for all state variables of a system to be constant, there must be a flow through the system.

Schematic of continuous processing

Biopharma process, starting from complete product: product moves



Assembly line, starting from product component: intermediate moves



PCC = periodic counter-current chromatography
STP = straight-through processing
API = active pharmaceutical ingredient

FDA perspective on continuous manufacturing

- Switch from batch to continuous production
- Required science is available
- There are no regulatory hurdles but a lack of knowledge

Regulatory definitions

- U.S. FDA definitions of both lot and batch in 21 CFR 210.3 are both applicable to continuous processes.
- The EMA has adopted the ICH Q7 definition: “In the case of continuous or semi-continuous production, a batch may correspond to a defined fraction of the production. The batch size can be defined either by a fixed quantity or by the amount produced in a fixed time interval.”

Other industries

*Blast furnaces were in use in China since the 1st century AD and reached the west more than 1,000 years later.
Processes began to operate continuously during the 19th century.
By early 20th century, continuous processes were common.

Other industries

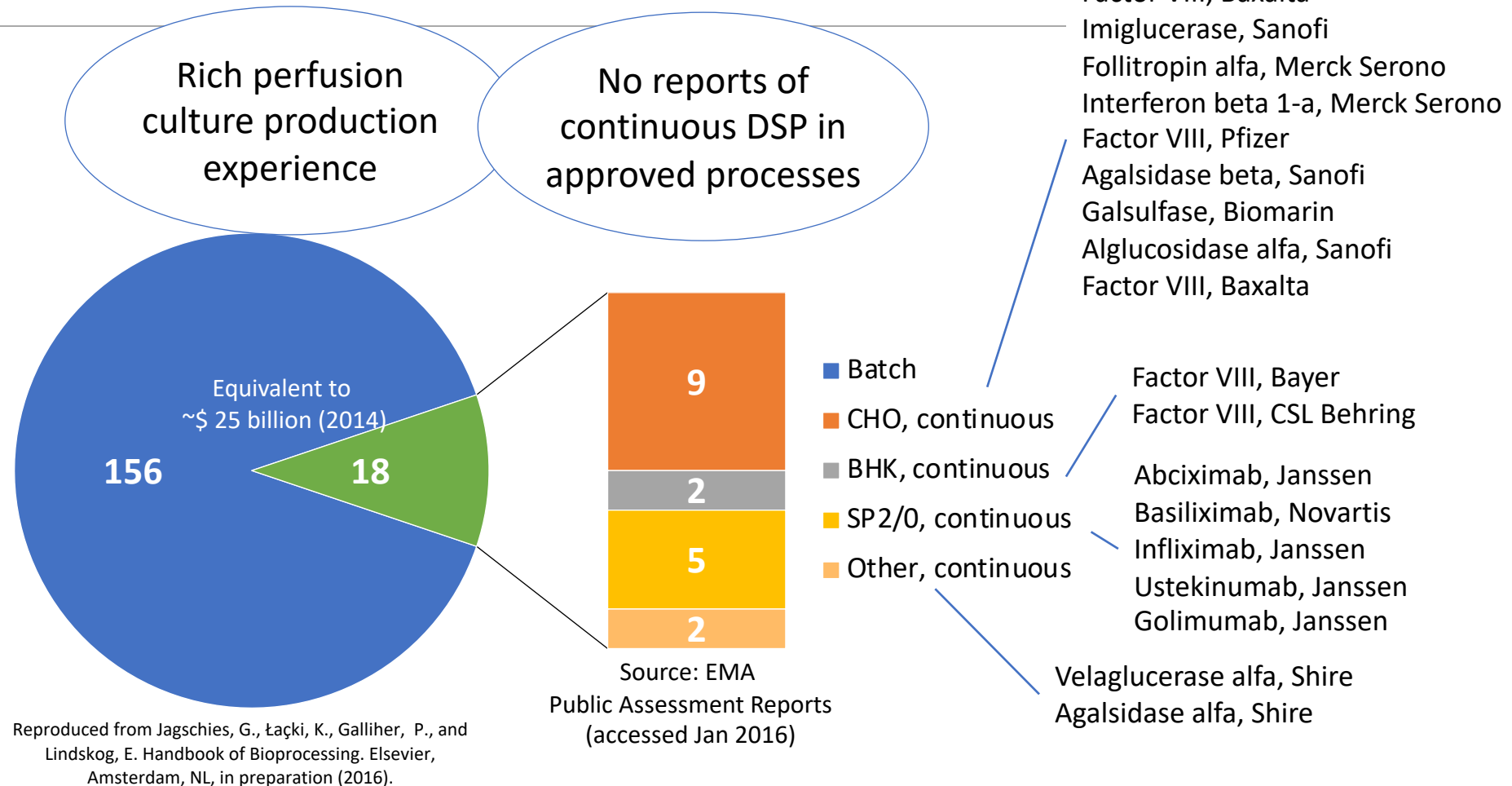
Why is it done?

- Mass production
- Low-margin products
- Enabling economics
- Consistent quality
- Smaller footprint
- Less labor

Continuous processes

- Iron: blast furnace*
- Steel: continuous casting
- Paper: the Fourdrinier machine
- Flour: flour milling
- Oil industry: continuous distillation
- Cement: rotary kiln
- Glass: float glass process
- Synthetic fibers: melt-spinning process
- Electricity: power plants
- Water: sewage or wastewater treatment
- Cars: assembly lines

Current approved injectables (APIs*)



Why is it done?

Two main reasons:
quality and cost

In other industries

- Mass production
- Low-margin products
- Enabling economics
- Consistent quality
- Smaller footprint
- Less labor

In biopharma

- Mass production
- Low-margin products
- Enabling economics
- Consistent quality
- Smaller footprint
- Less labor

At first glance, biopharma does not qualify: not on mass and certainly not on low margin.

Rules of technology selection

Novel tested technologies not used ...

- No significant or sufficient advantage in either performance or economics.
- Regulatory hurdles higher than for conventional methods.
- Yielded product has insufficient quality or quantity.
- Inability to scale up.
- High cost associated with the alternative.
- Logistics of manufacturing: custom equipment, supply chain of consumables.
- Poor fit with existing processes or facilities.
- Method not well understood or appreciated.

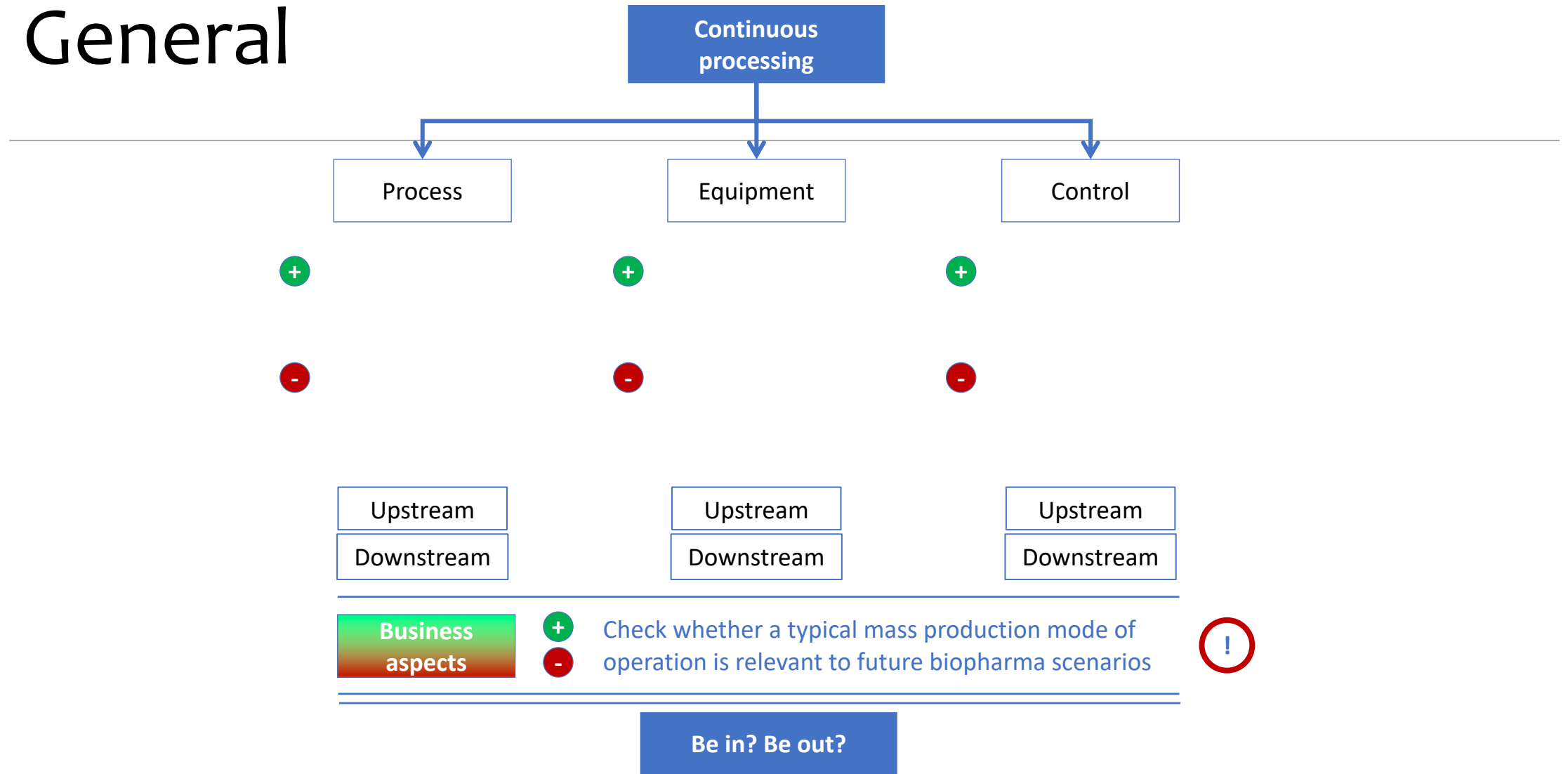
Most alternatives address "multiton", not "multiproduct".

Large improvement margin required.

Risk averse management approach to new technology.

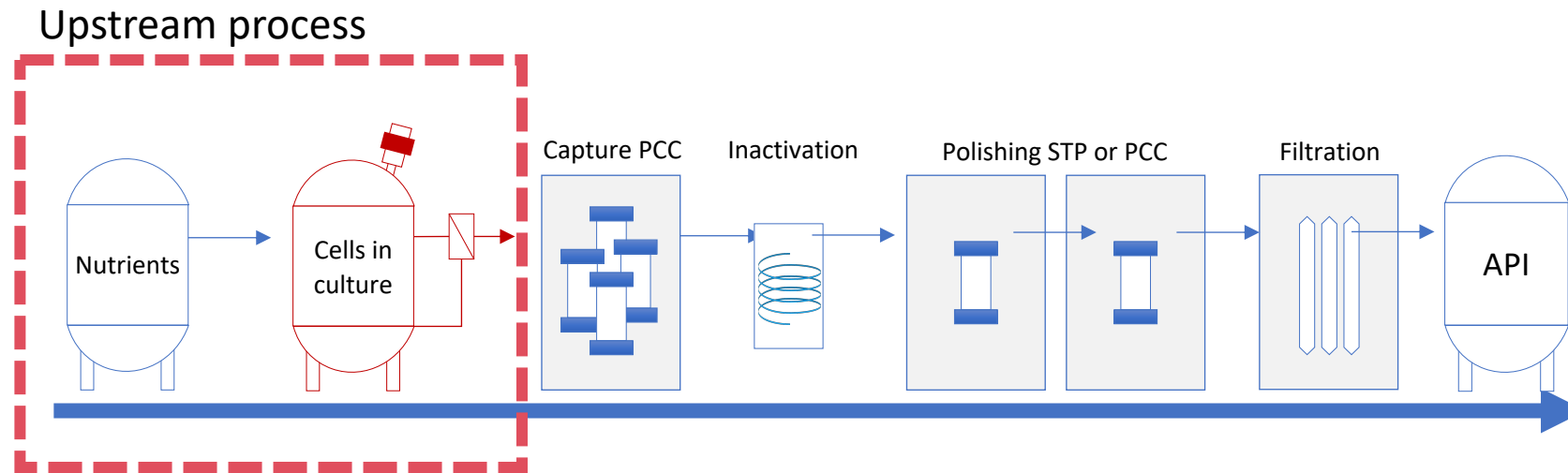
Tran, R., Lacki, K., Davidson, A., Sharma, B., Titchener-Hooker, N. Changing manufacturing paradigms in downstream processing and the role of alternative bioseparation technologies. *Journal of Chemical Technology and Biotechnology*, Volume 89, Issue 10, pages 1534–1544, October 2014

General



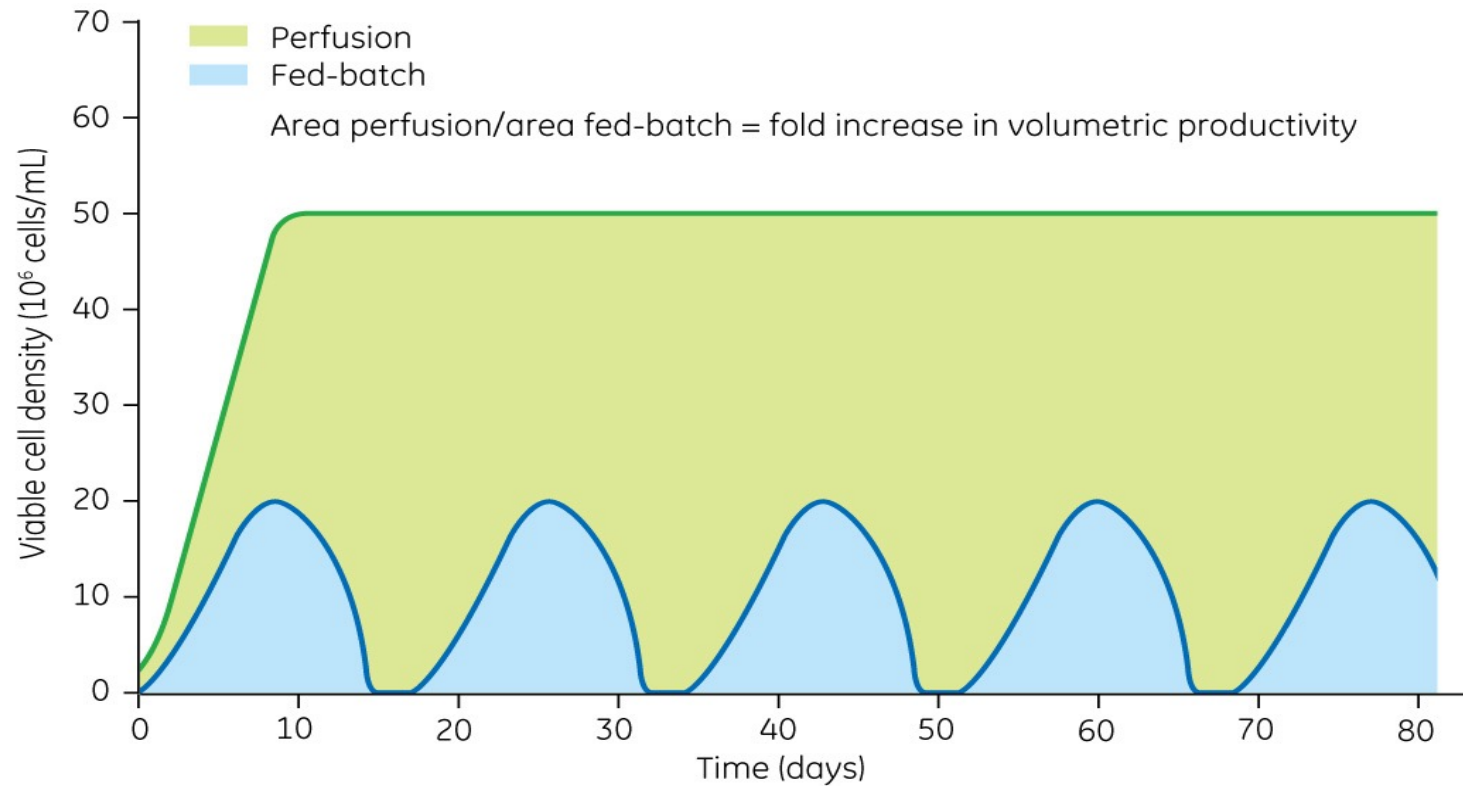
Upstream processing aspects

Process overview

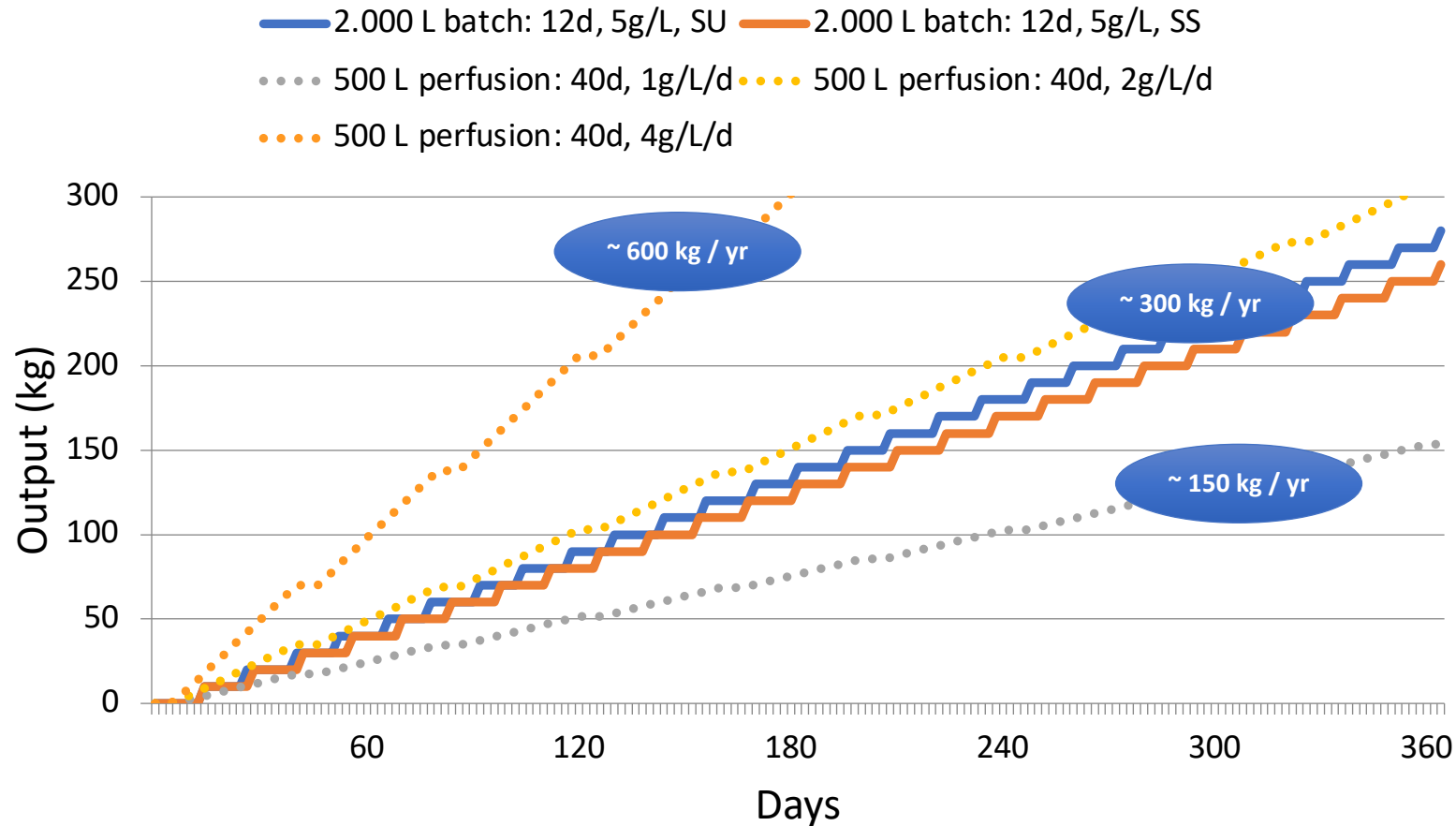


PCC = periodic counter-current chromatography
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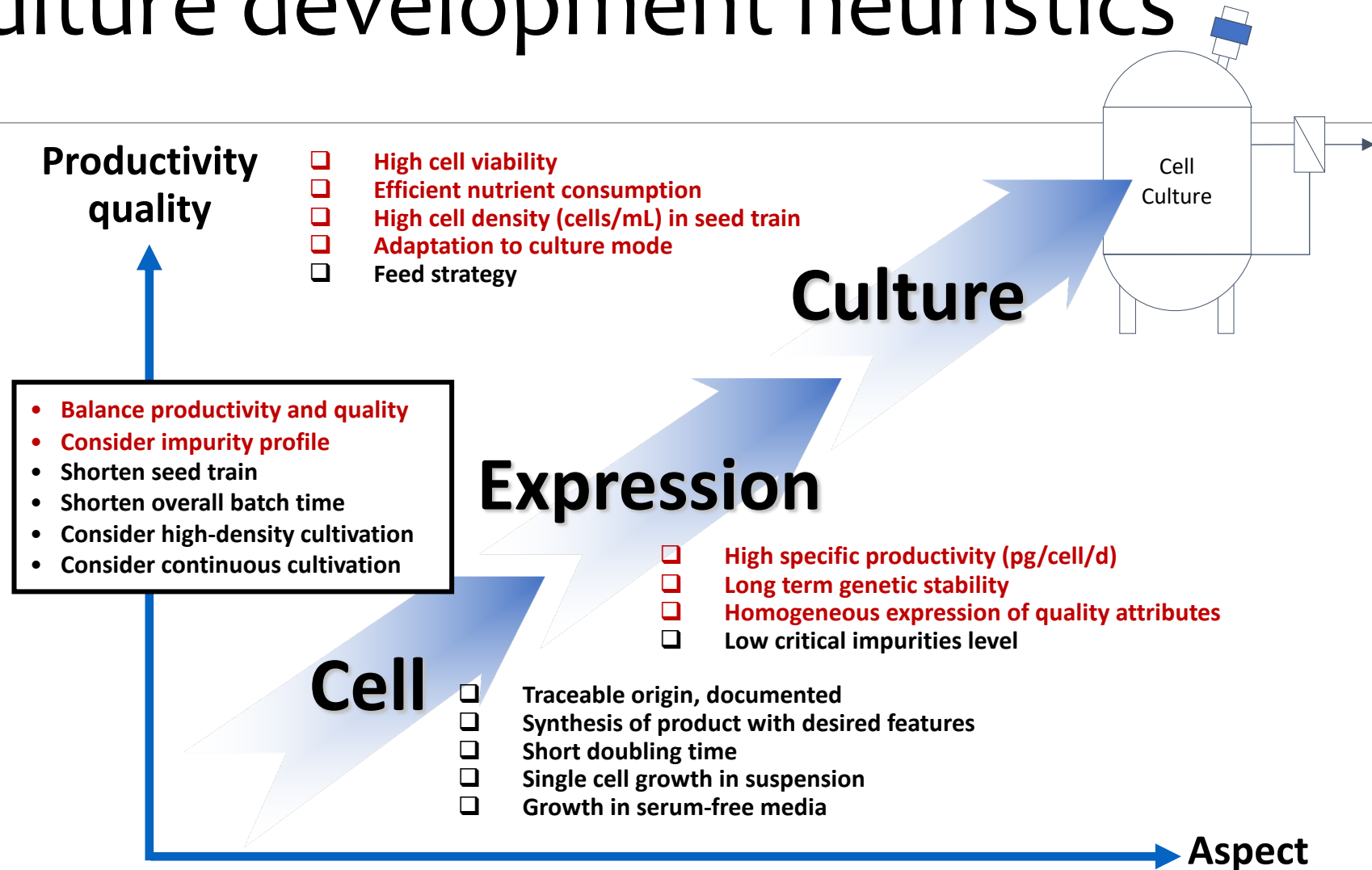
Illustration of volumetric productivity increase for fed-batch vs perfusion



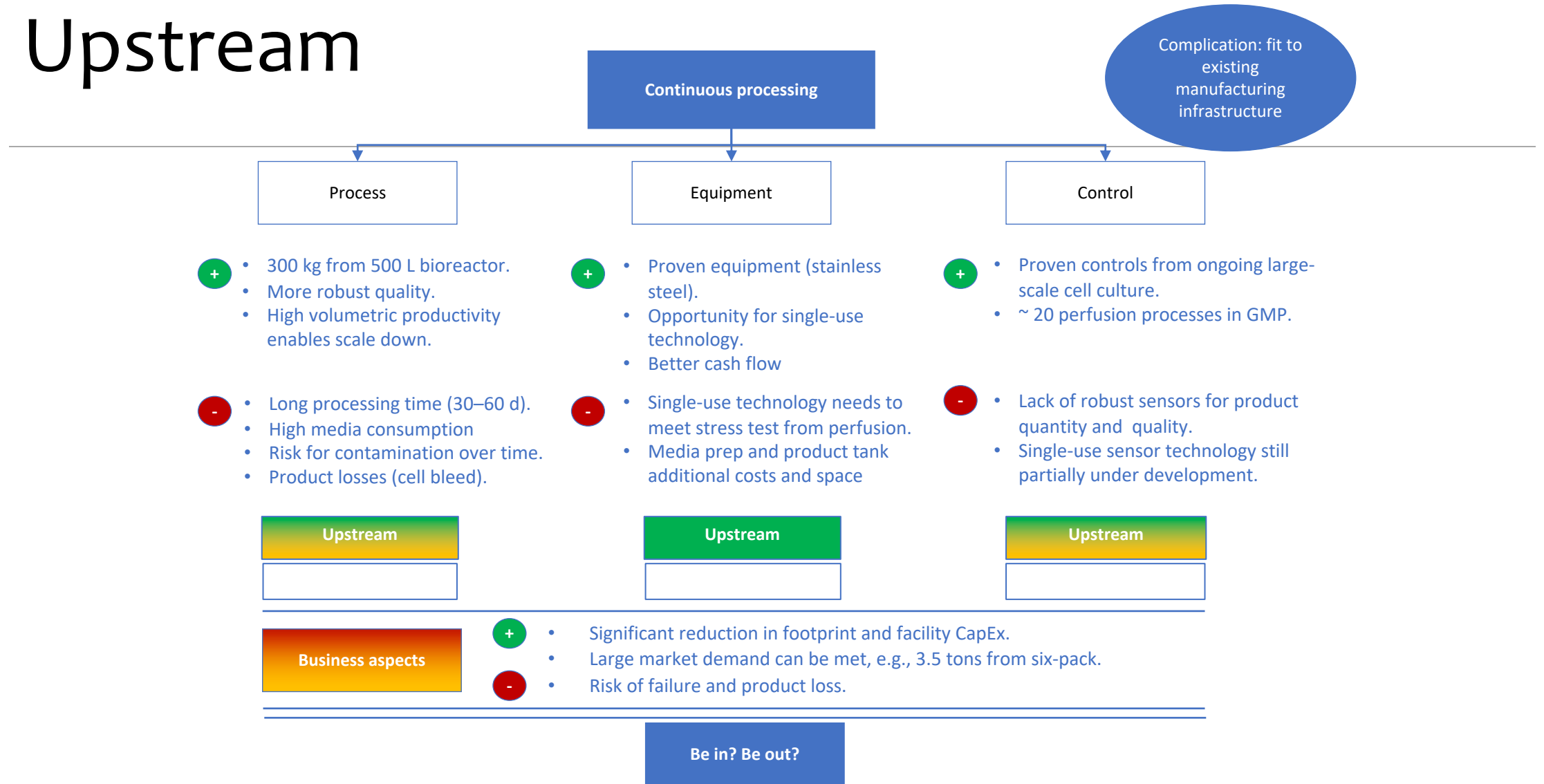
500 L perfusion bioreactor at 2 g/L/d equiv. to 2.000 L bioreactor at 5g/L/batch (~ 300 kg/yr)



Cell culture development heuristics



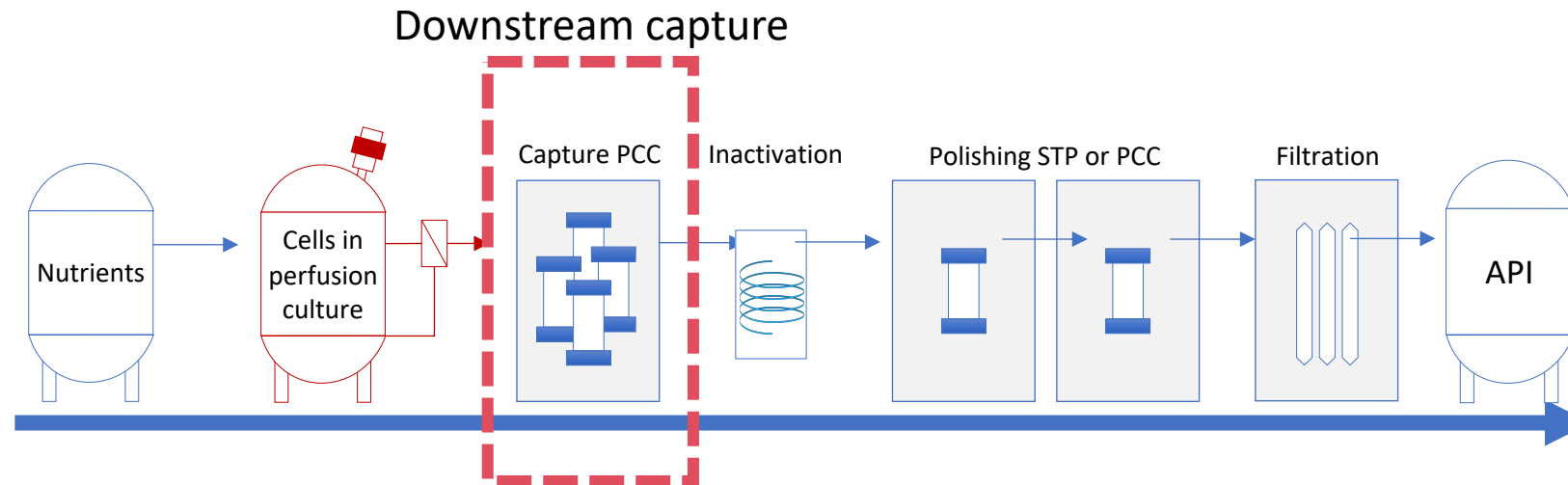
Upstream



CapEx = capital expenditure

Downstream processing aspects

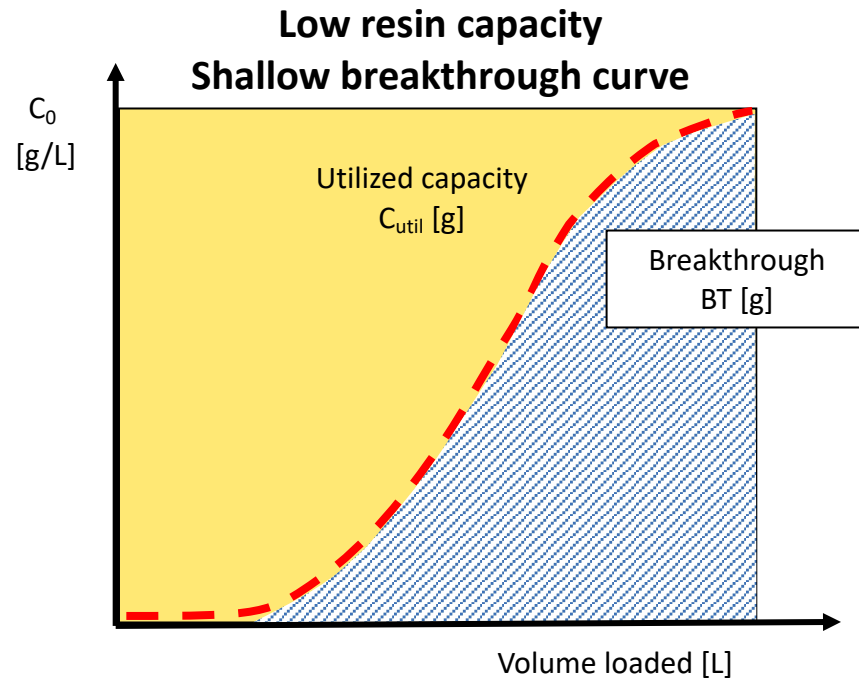
Process overview



PCC = periodic counter-current chromatography
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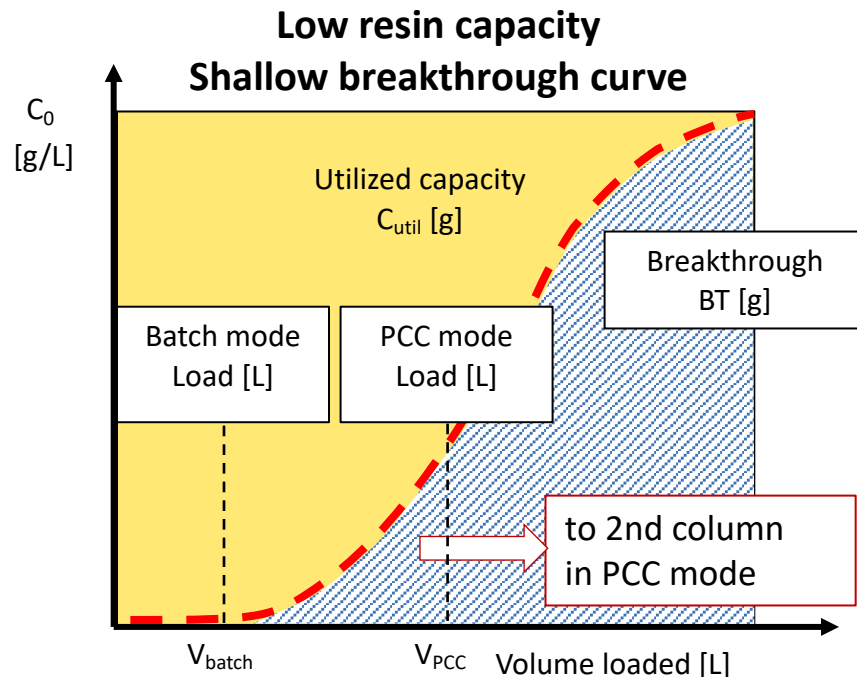
Understanding PCC* Features & Benefits

*Periodic Countercurrent Chromatography



Understanding PCC* Features & Benefits

*Periodic Countercurrent Chromatography



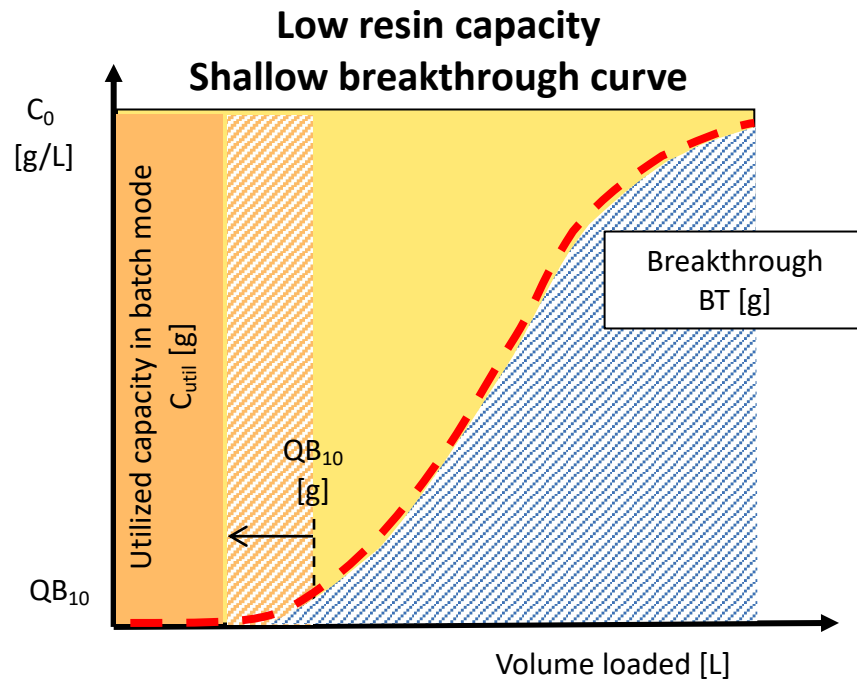
V_{PCC}/V_{batch} describes the potential for PCC mode to achieve a high % increase in utilized capacity over capacity in batch mode.

The breakthrough is bound to the next column in the PCC sequence.

Once the system has gone through the first cycle, all columns are loaded with $\sim V_{PCC}$ minus the volume equivalent to the breakthrough product already bound.

Understanding PCC* Features & Benefits

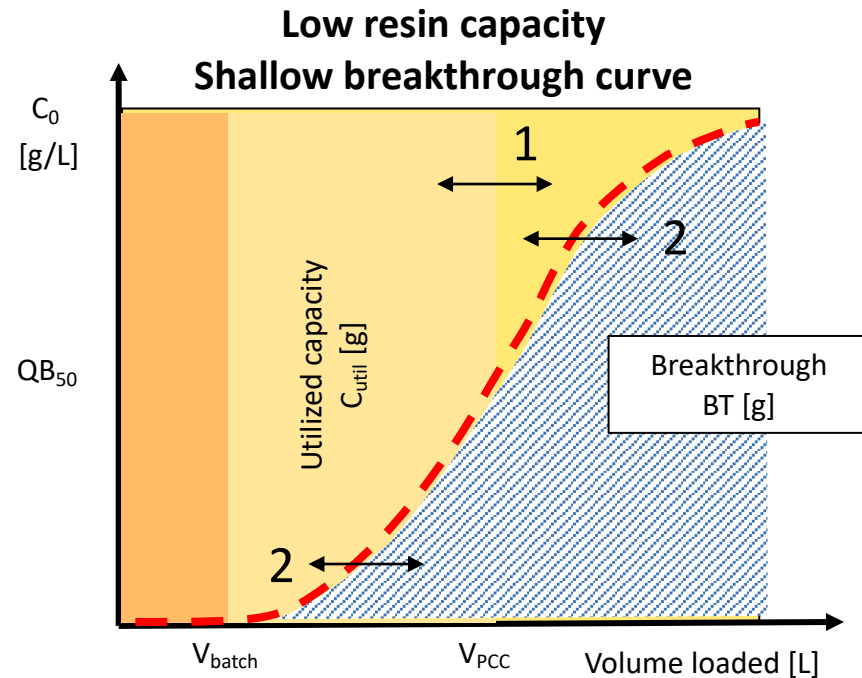
*Periodic Countercurrent Chromatography



In batch mode, the utilized capacity usually is set to Q_{B10} minus a safety factor depending on the steepness of the breakthrough curve and the expected loss of capacity over the resin lifetime.

Understanding PCC* Features & Benefits

*Periodic Countercurrent Chromatography

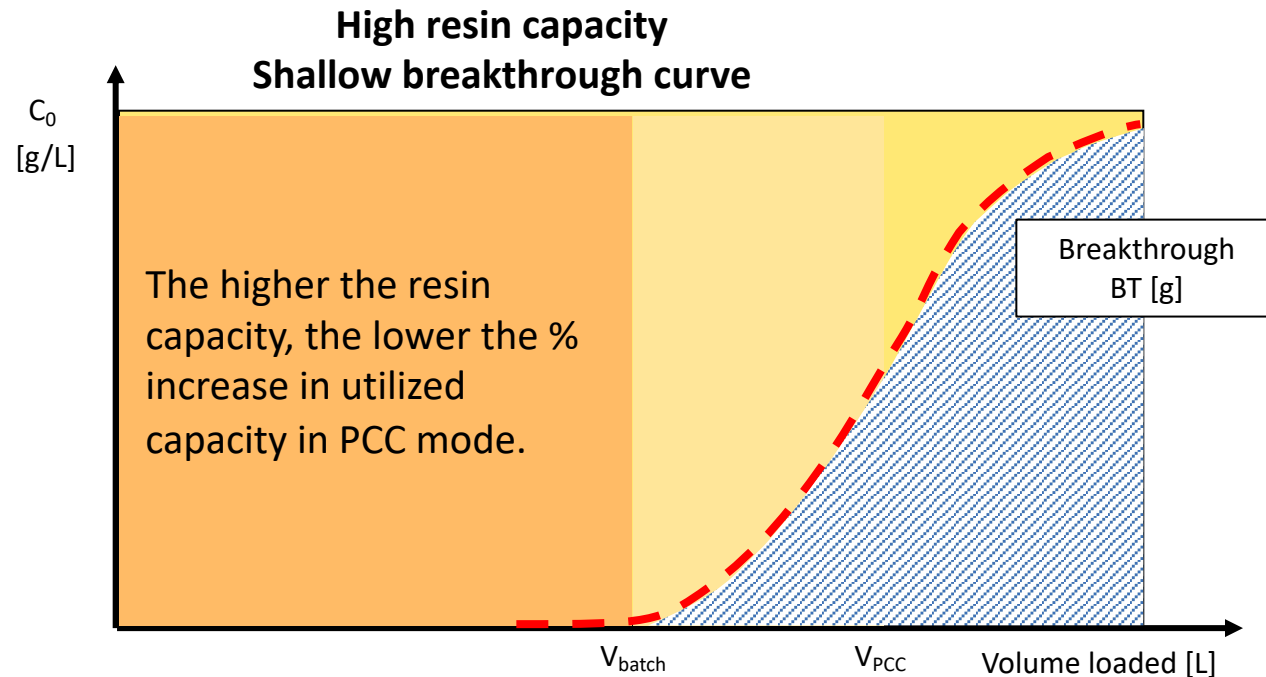


In PCC mode, the overload of the column can be varied by:

1. Higher / lower load volume at constant residence time
2. Shorter / longer residence time — change of breakthrough curve

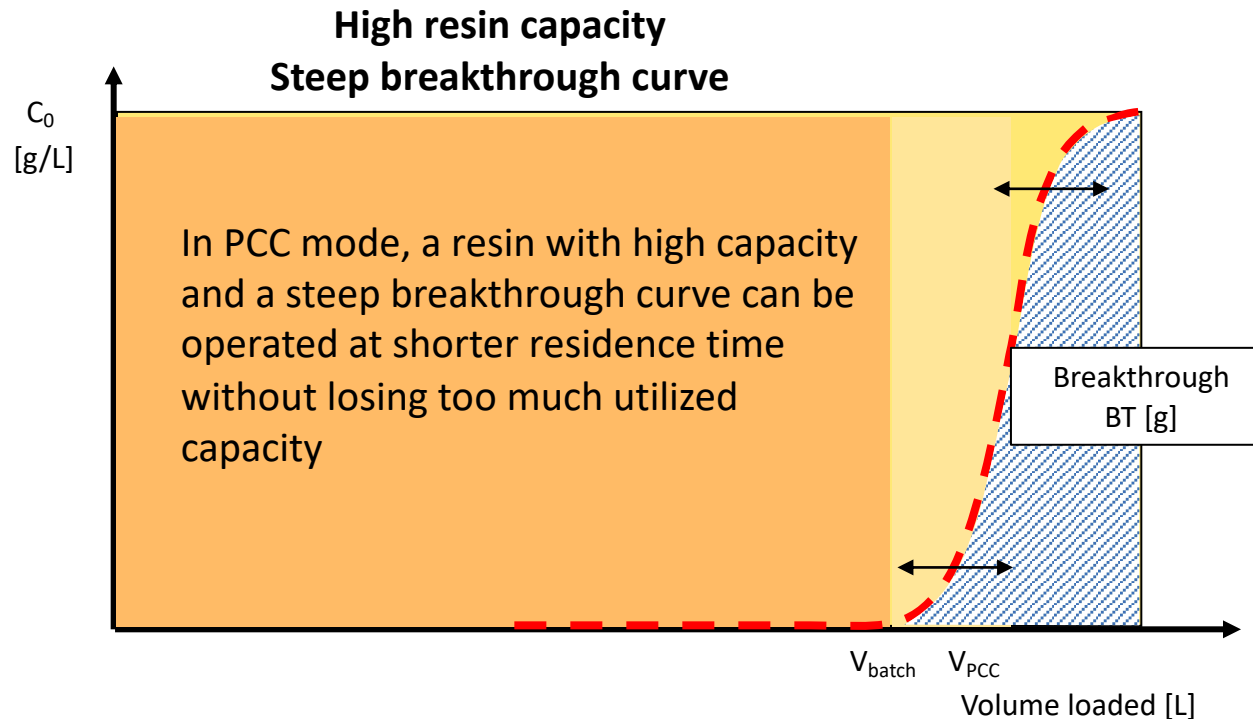
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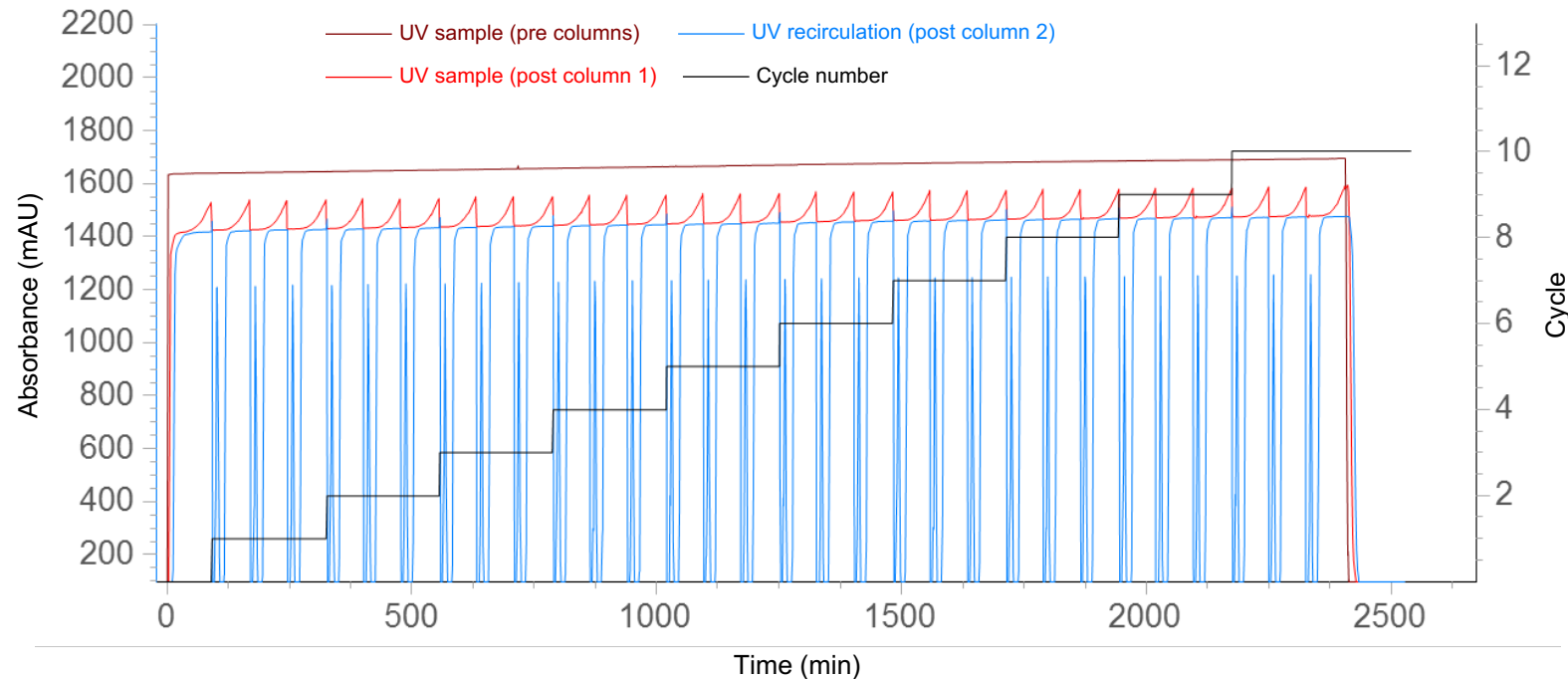


Understanding PCC* Features & Benefits

*Periodic Countercurrent Chromatography



Chromatograms from ten cycles on three columns (total of 30 loadings)



67 g/L loaded onto each column compared with loading 43 g/L in batch mode gives a 56% increase in capacity utilization.

Resin consumption with PCC

Opportunities

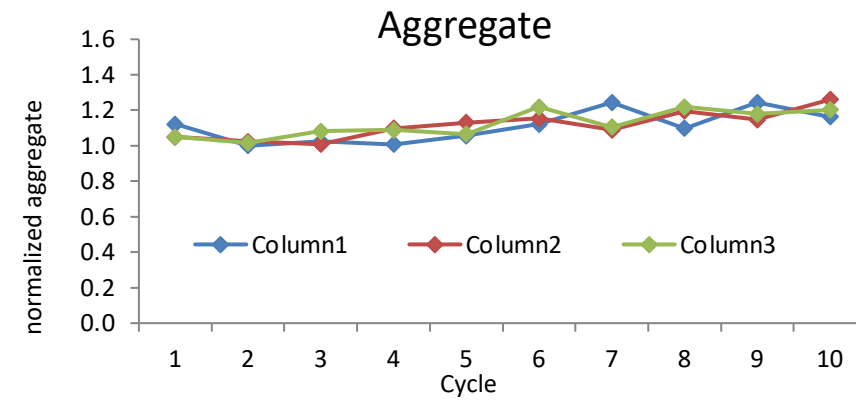
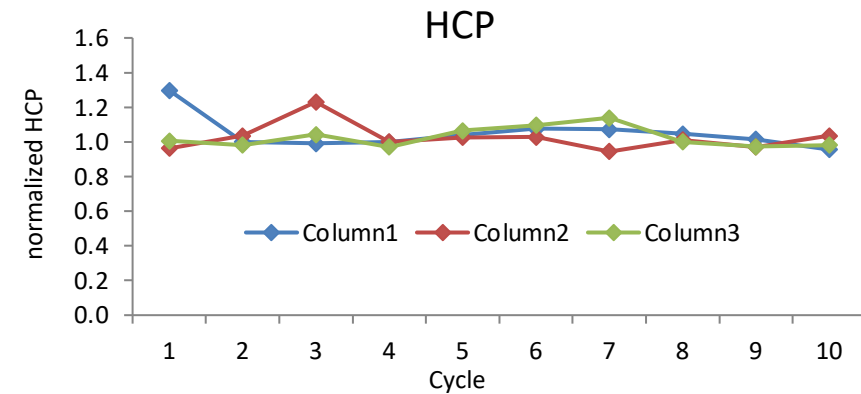
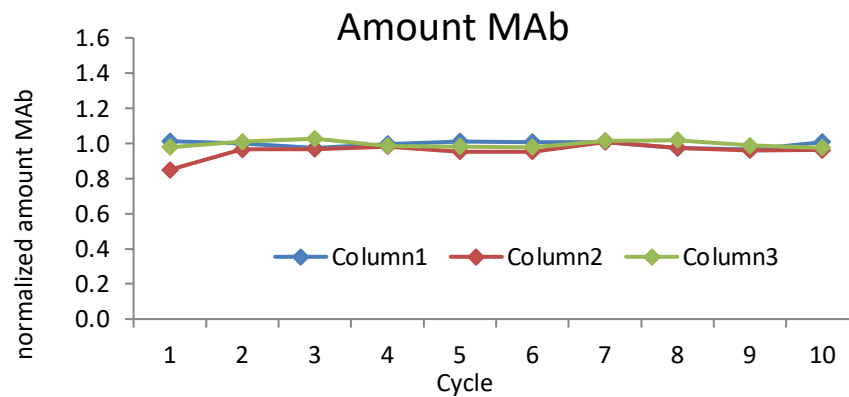
- The use of PCC eliminates the safety factor for loading and gains 20-30% of capacity this way
- The resin can be overloaded up to 50% breakthrough, which provides an additional gain of 20-30% capacity
- Load and elution happen in parallel and lead to productivity gains
- Columns can be run at short residence time with large number of cycles per reactor batch
- Limited Mab quantities can be made with full resin lifetime utilization

Limitations

- Running in smaller columns and allowing a breakthrough does not reduce total resin consumption for a given Mab quantity assuming lifetime utilization is the same.
- Running faster reduces the capacity to varying degrees
- Running step at higher productivity does not necessarily have an effect on the overall process
- Smaller columns reach their lifetime earlier, may lead to additional replacement costs

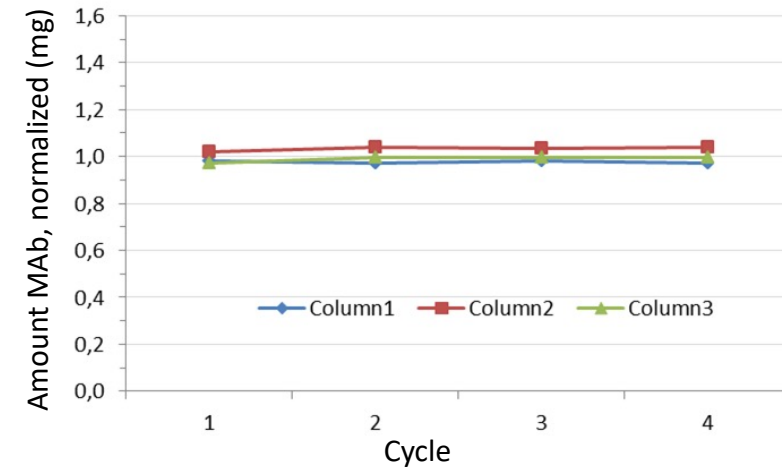
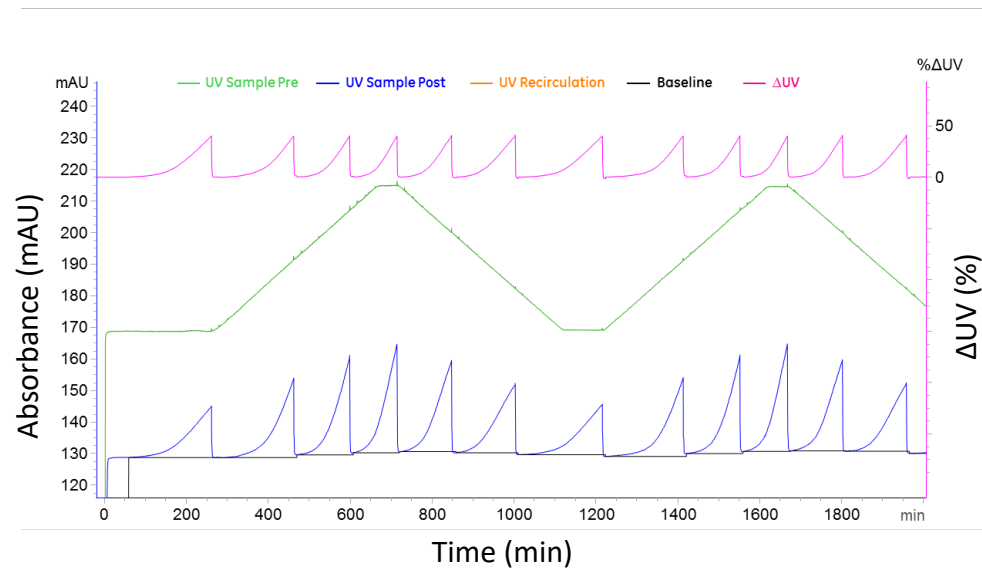
Multiple cycles with consistent performance

- Robust performance over 10 cycles.
- Minimal variation in MAb amount, aggregate content, and host cell protein (HCP) content between columns.

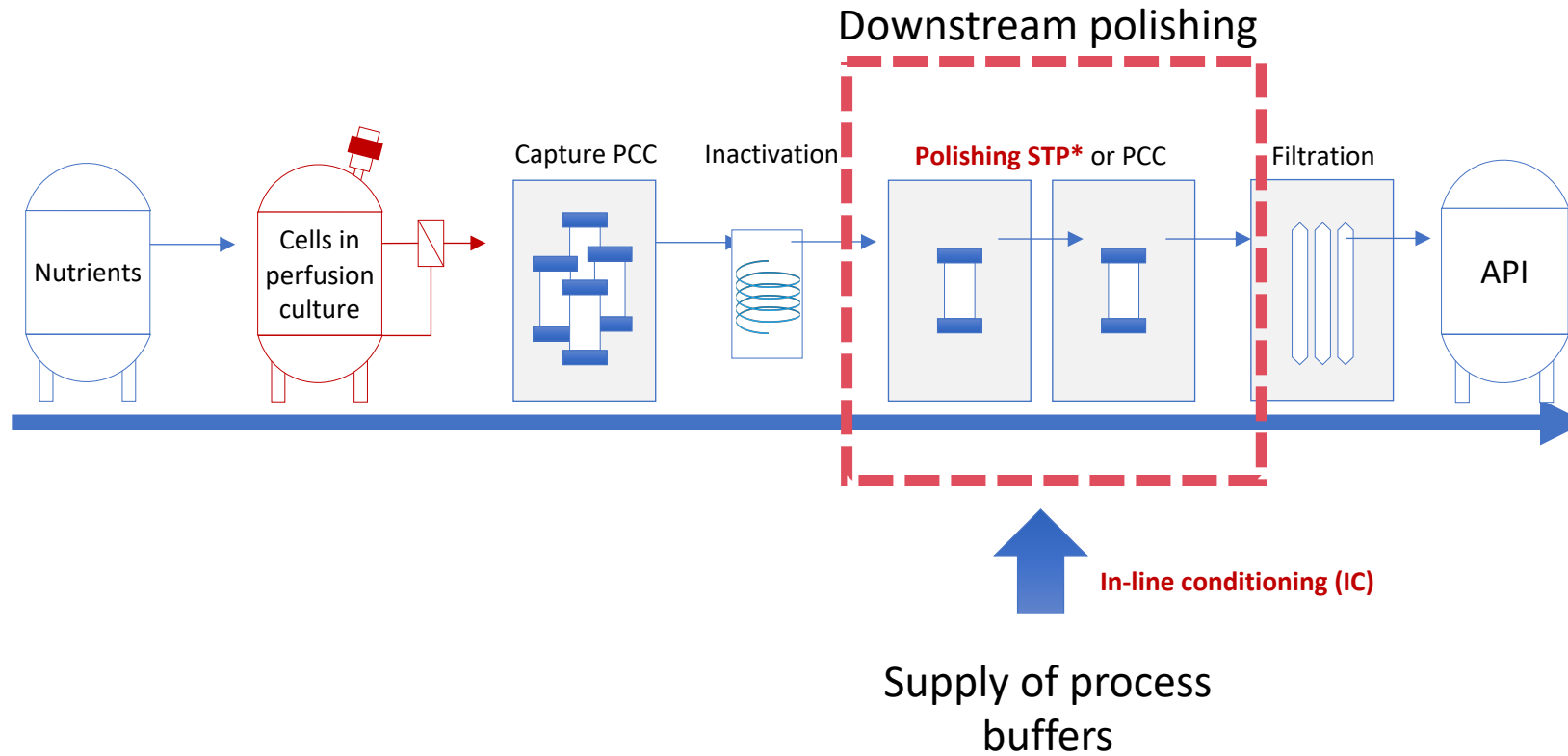


Demonstration of dynamic control: variable sample concentration

- The dynamic control was challenged by performing a feed gradient run from 1 to 2 g IgG/L with BSA as background.
- The breakthrough curves follow the gradient change closely.

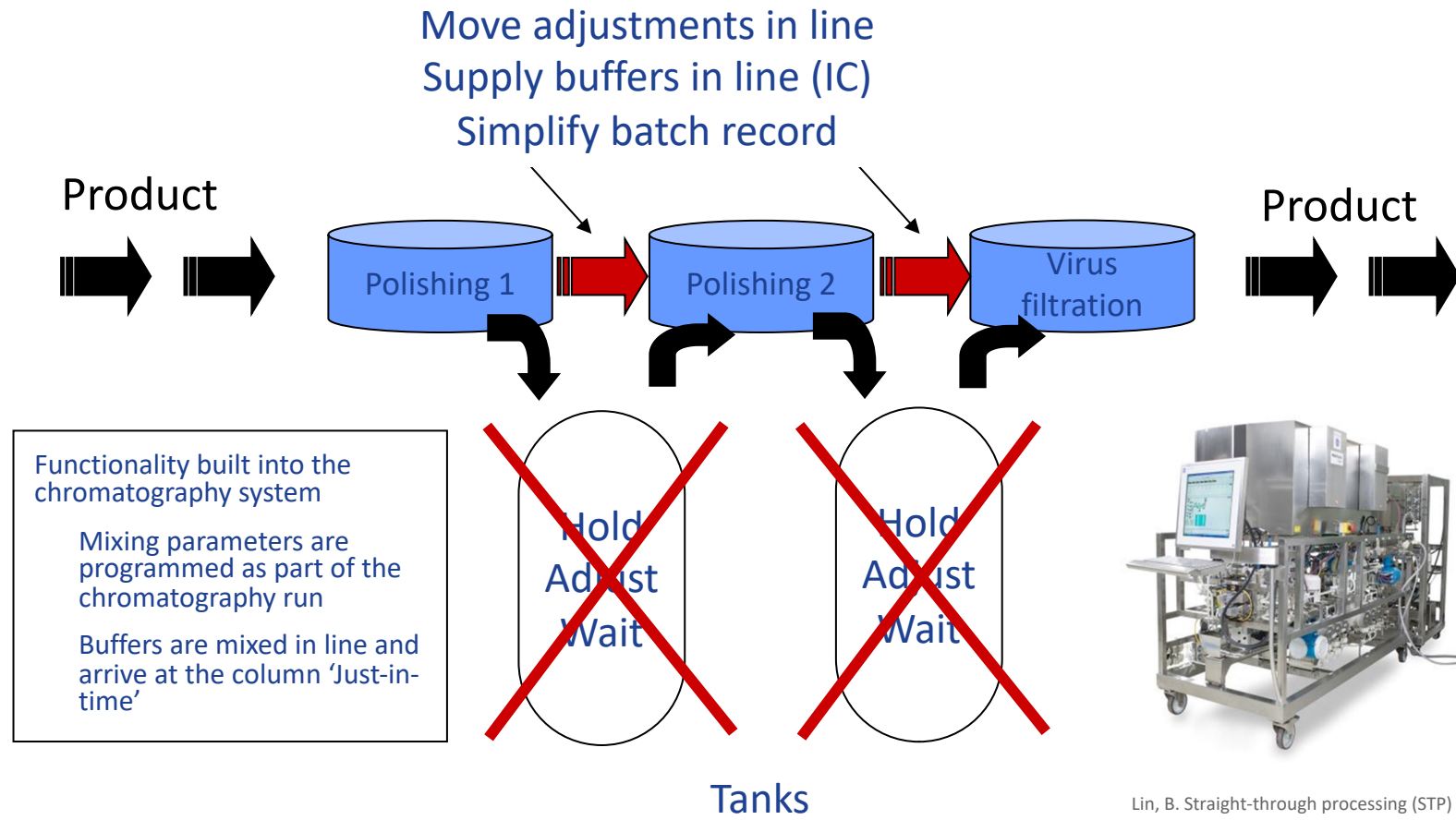


Process overview



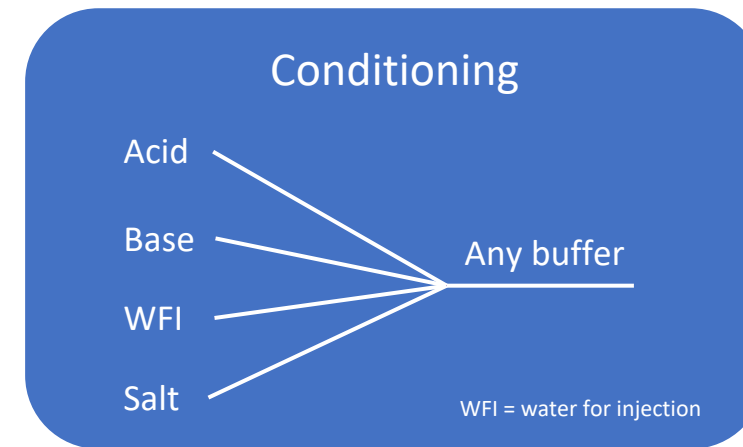
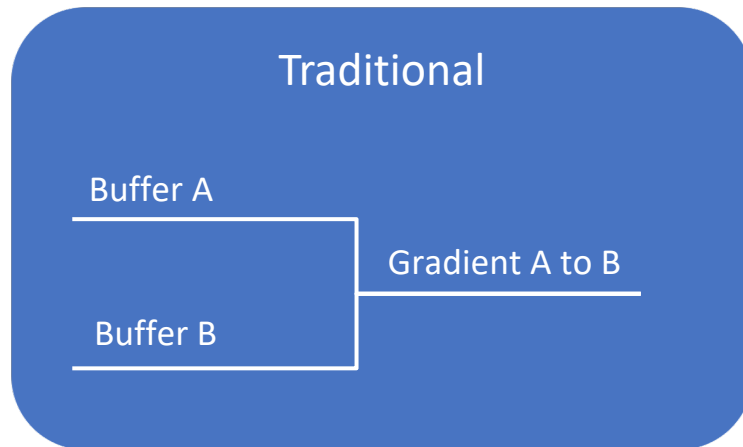
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Straight-through processing (STP) combined with in-line conditioning (IC) for buffers

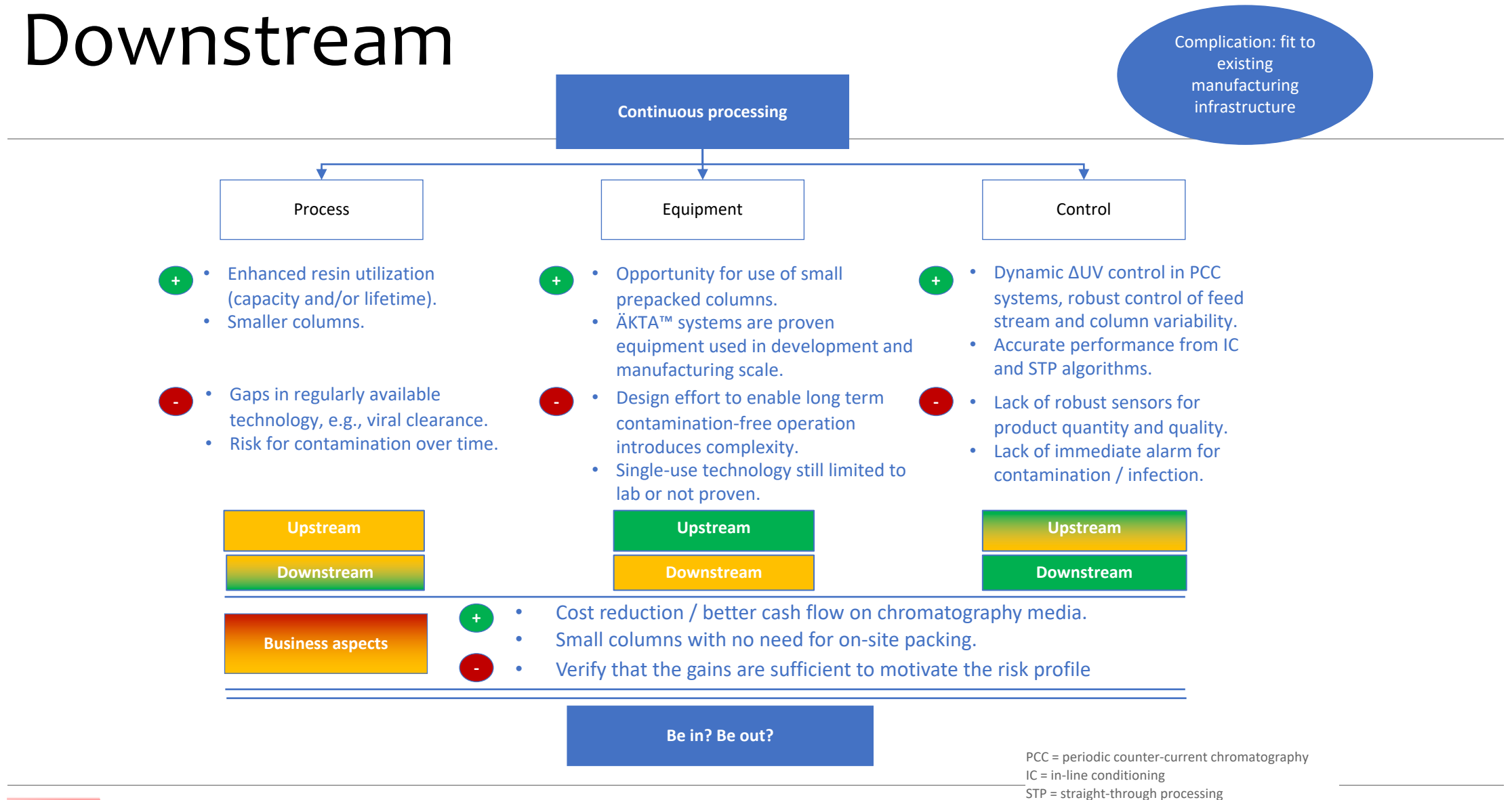


Basics of IC buffer preparation

- The traditional approach is to prepare and mix each buffer, in the exact volume needed, according to specifications.
- With IC, the individual components of the buffers are separated and concentrated. These component concentrates are later mixed with each other and water to virtually any buffer strength, pH, or conductivity specification. The same concentrate is used to create multiple buffers.

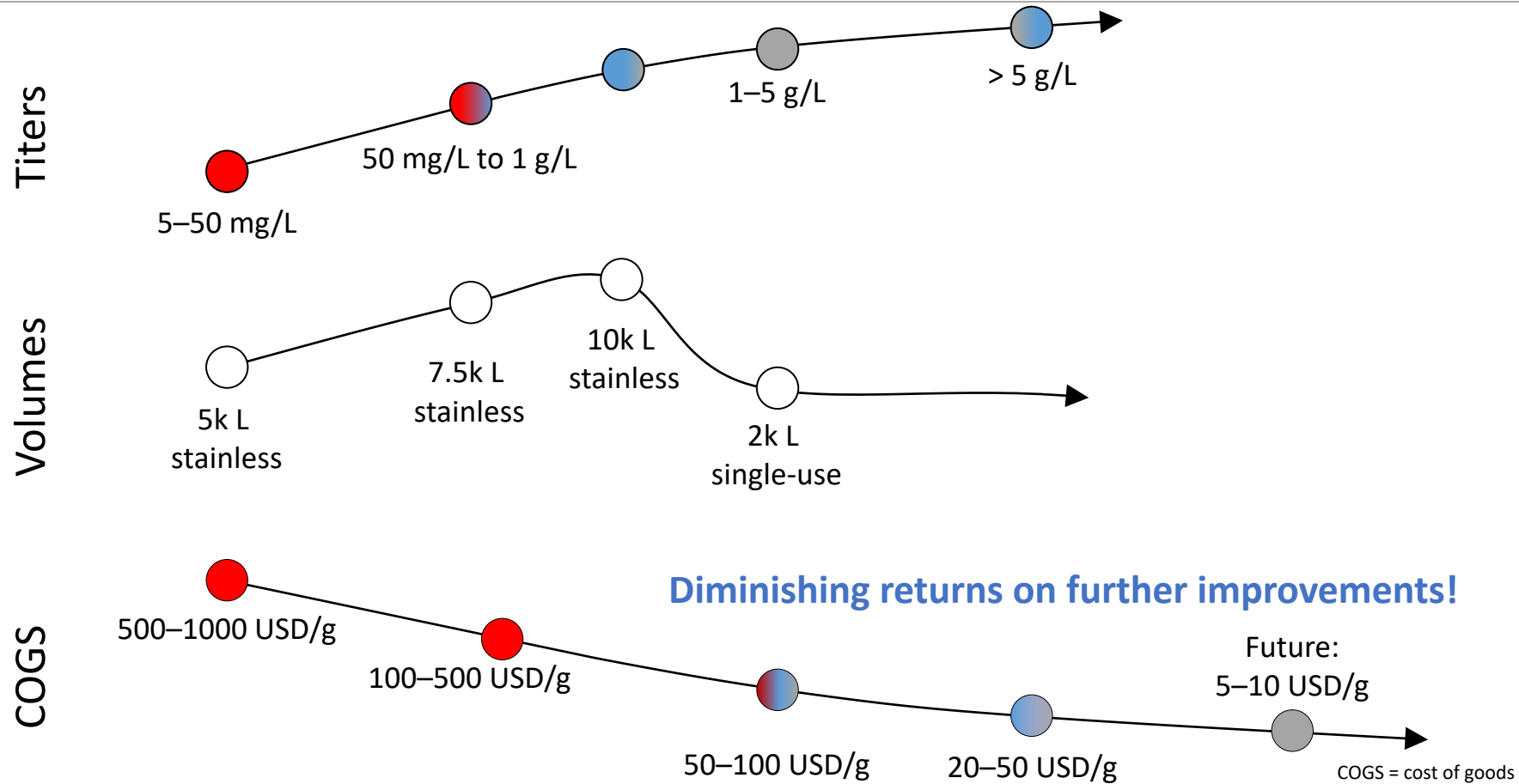


Downstream



Economic considerations

Antibody production: increasing in productivity, decreasing in scale, pressure on COGS



How does the gain in volumetric productivities translate into improved process economics?

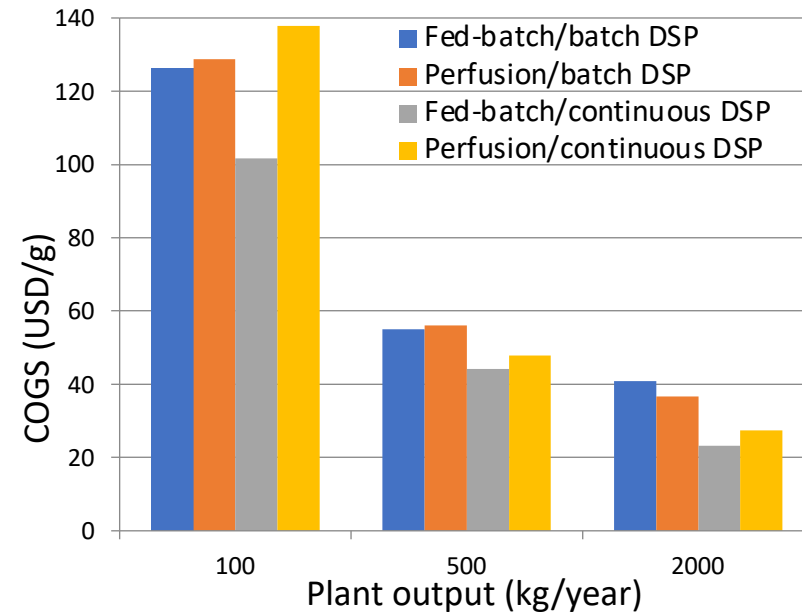
- Different approaches to describe the business model and facilitate decision making have been published
 - UCL^{1,2}
 - BioPharma Services¹
 - Genzyme³

¹ Pollock J. *et al.*, Fed-batch and perfusion culture processes: economic, environmental, and operational feasibility under uncertainty, *Biotechnol. Bioeng.* **110**, 206–219 (2013)

² Pollock J. *et al.*, Optimising the design and operation of semi-continuous affinity chromatography for clinical and commercial manufacture, *J. Chromatogr. A*, **1284**, 17–27 (2013)

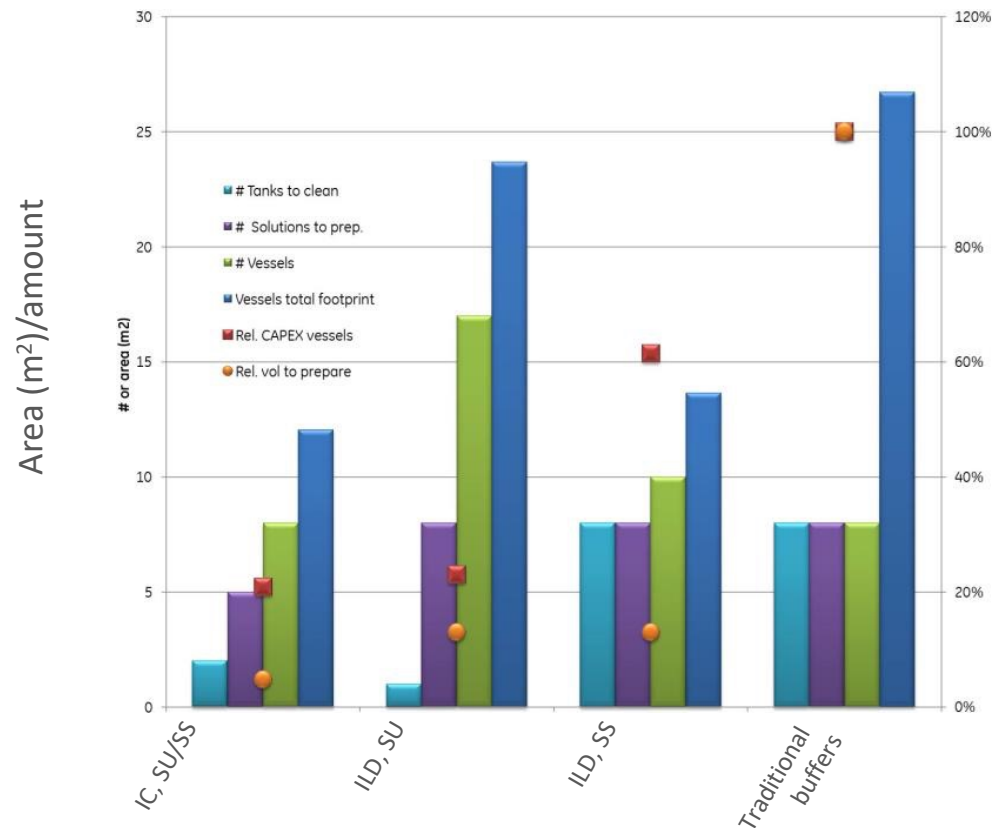
³ Walther J. *et al.*, The business impact of an integrated continuous biomanufacturing platform for recombinant protein production, *J. Biotechnol.*, **213**, 3–12 (2015)

- Process economy comparison, batch vs fed-batch vs continuous processing



COGS = cost of goods, DSP = downstream processing, Data courtesy of BioPharm Services, 2015

Improved process economy for IC compared with traditional buffer preparation



- Comparing buffer hold vessel needs in four scenarios:
 - IC with single-use (SU) vessels (all except acid and WFI hold)
 - ILD with single-use vessels
 - ILD with stainless steel (SS) vessels
 - Traditional buffers
- Four chromatography steps in an albumin purification process
- 50% footprint reduction
- 38% less volume required

CAPEX = capital expenditures

Conclusions:

Are you in or are you out?

Conclusion

Be in?

- Large demand
- Competition, price pressure
- Build new and small, low CapEx
- Technology proven to work
- Development effort similar
- Risks are manageable
- Lowest cost, best cash flow
- Implement all the way...?

Be out?

- Many small products
- Current margins preserved
- Fit into existing facility
- Contamination, other failure
- Development more complex
- Risks are too high
- Gains are diminishing
- Implement certain steps...?